Crypto for SPDM: From Legacy to Quantum-Resistance and More

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- The information in this presentation represents a snapshot of work in progress within the DMTF SPDM WG.
- This information is subject to change without notice. The standard specifications remain the normative reference for all information.
- For additional information, see the DMTF website.
- This information is a summary of the information that will appear in the specifications. See the specifications for further details.

Why Revisit SPDM's Cryptography Selection and Integration?

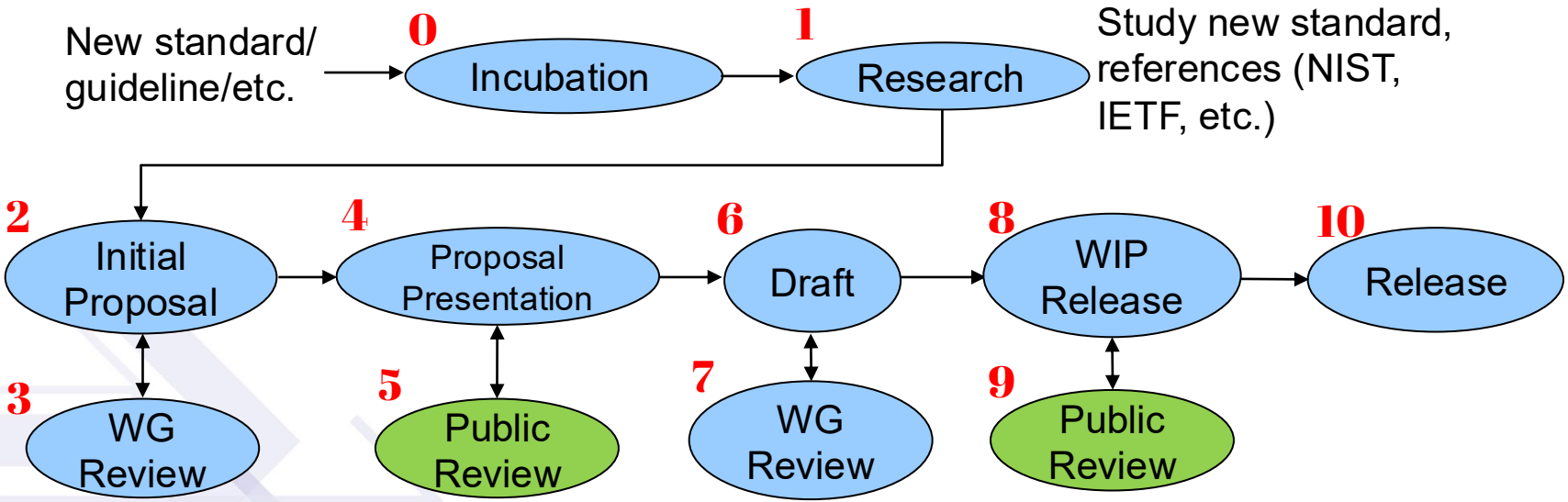
- Bring transparency to algorithm selection process
- Understand on design considerations
- How to get involved
- Increase feedback and participation

The Crypto Evolvement: SPDM 1.0 → 1.5

Legacy
PQC
Hybrid

Algorithm	1.0 12/2019	1.1 7/2020	1.2 12/2021	1.3 6/2023	1.4 5/2025	1.5 Q1/2027
AEAD	No support	AES-GCM-128/256; Chacha20-Poly1305	+ SM4-GCM			
Hash	SHA2/3-256/384/512		+ SM3			
Key exchange	No support	ffdhe2048/3072/4096; secp256r1/384r1/521r1	+ SM2		+ ML-KEM-512/768/1024	+ MLKEM1024-ECDH-P384; MLKEM1024-X448; MLKEM1024-ECDH-P521
Signature	RSA-SSA/PSS-2048/3072/4096; ECDSA-P256/384/521		+ SM2; Ed25519/448		+ ML-DSA-44/65/87; SLH-DSA-SHA2/SHAKE-128s/f/192s/f/256s/f	+ MLDSA87-ECDSA-P384; MLDSA87-Ed448; MLDSA87-RSAPSS3072/4096

SPDM Crypto Design Process



SPDM 1.5 – Hybrid: Incubation and Research

- Incubation: Authority/Industry recommendation/requirement
  - ANSSI views on the Post-Quantum Cryptography transition [link]
  - BSI: Quantum Technologies and Quantum-Safe Cryptography [link]
- Research: IETF RFCs / Drafts
  - Composite Module-Lattice-Based Digital Signature Algorithm (ML-DSA) for use in X.509 Public Key Infrastructure [link]
    - For algorithm combinations
  - Composite ML-KEM for use in X.509 Public Key Infrastructure [link]
    - For algorithm combinations
  - Hybrid Key Exchange in TLS 1.3 [link]
    - For shared secret combiner in key schedule

SPDM 1.5 – Hybrid: Initial Proposal

- **Minimalist Approach:**
  - Hybrid is a transitional feature so hybrid design should be as simple as possible. Only a small number of hybrid combinations are supported to facilitate compatibility and reduce cost.
  - Reuse pre-existing messages and SPDM-defined functions such as SpdmSign/Verify and key schedule to minimize code changes.
- **Upgradability**:
  - SPDM 1.4 endpoints shall be able to support hybrid via firmware update by reusing pre-existing legacy and PQC certificate chains.
- **Flexibility:**
  - SPDM 1.5 does not preclude supporting composite certificates defined in IETF Drafts

SPDM 1.5 – Hybrid: Draft - Design Principles (1/3)

- Message fields related to certificate chains, certificate signing requests (CSR), or digests may refer to one or two certificate chains
- When referring to two certificate chains: concatenation of PQC and legacy certificates, CSRs, and digests
- Order of concatenation matches order in the hybrid algorithm name
- MLDSA87-ECDSA-P384 implies the following:
  - Concat(ML-DSA-87 certificate chain/CSR/digest, ECDSA-P384 certificate chain/CSR/digest)

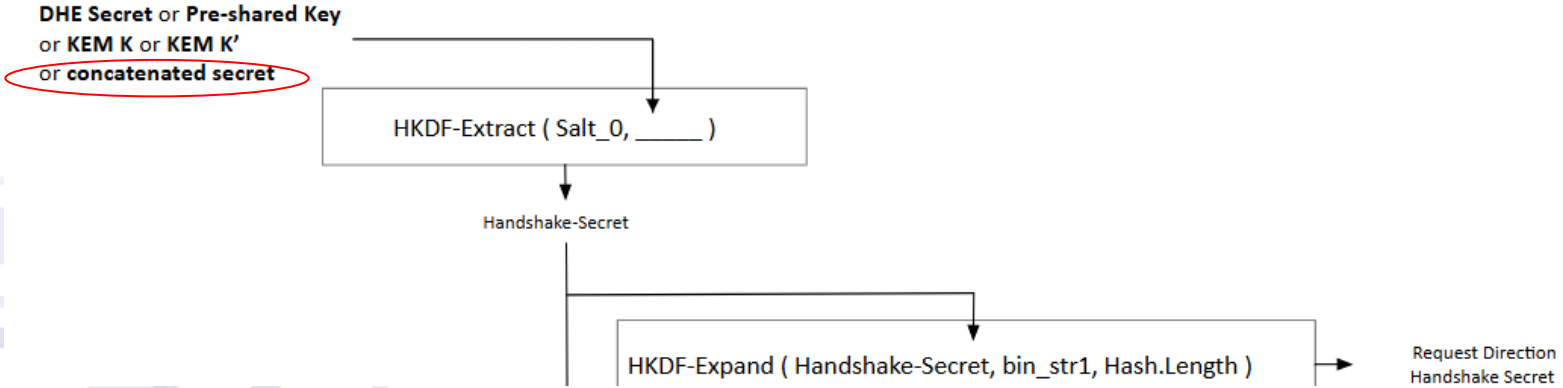
SPDM 1.5 – Hybrid: Draft - Design Principles (2/3)

- Message fields that contain data of signature, ciphertext (in key exchange), or asymmetric public key (in key exchange), will contain the concatenation of two pieces of data for the legacy and PQC algorithms, respectively.

2	AlgSupported	`HybridAsymAlgCount`	Shall be the bit mask listing Requester-supported SPDM-enumerated hybrid asymmetric key signature algorithms for the purpose of signature verification by the Requester. The bit mask shall be as defined in this clause. If the Requester does not support any request/response pair that requires signature verification, this value shall be set to zero. If the Requester will not send any requests that require a signature, this value shall be set to zero. Let `SigLen` be the size of the signature in bytes. • Byte 0 Bit 0. MLDSA87-ECDSA-P384 where `SigLen` =4723. • Byte 0 Bit 1. MLDSA87-Ed448 where `SigLen` =4741. • Byte 0 Bit 2. MLDSA87-RSA3072-PSS where `SigLen` =5011. • Byte 0 Bit 3. MLDSA87-RSA4096-PSS where `SigLen` =5139. • All other values reserved.

SPDM 1.5 – Hybrid: Draft - Design Principles (3/3)

- Key exchange: the input to SPDM Key Schedule will be the concatenation of the two shared secrets resulted from Traditional and PQC, respectively.



Forward Looking

- Monitor NIST's progress of new PQC standardization, such as Falcon/FIPS206/FN-DSA and HQC.
- Monitor PQC standardization of other authorities, such as China's State Cryptography Administration.
- Adopt new PQC algorithms and/or hybrid combinations in future SPDM revisions if industry demands.
- Monitor hybrid guidelines from various authorities and remove hybrid support in a future SPDM revision if no longer required.

Call to Action

- Keep an eye out for feedback request for algorithm selection (https://www.dmtf.org/news)
- Review the latest WIP hybrid support (https://github.com/DMTF/SPDM-WG)
- Provide feedback to DMTF SPDM (https://www.dmtf.org/standards/feedback)



For more information,
visit dmtf.org

Learn about the SPDM working group at
dmtf.org/standards/spdm

Thank you!

What is SPDM?

- The DMTF's Security Protocols and Data Models (SPDM) Working group is responsible for the SPDM suite of specifications.
- These standards enable authentication, attestation and key exchange to assist in enabling infrastructure security.
- Other specifications define enhancements to infrastructure security through additional functionality or mapping SPDM to additional transports.

